



Your Section Title

An optional subtitle or tagline



Lectures



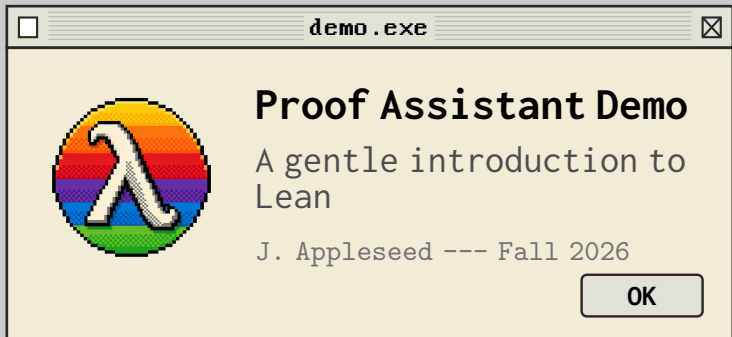
cats



README



Exercises





LEAN: Proof Assistant Demo Template

- Structural Example: Definition
- Key Axioms & Lemmas
- Theorem Application Steps
- Formalization Discussion

```
Lean Editor.pro
Monaco
1 theorem add_assoc (n m k : N) : n + m + k = n + (m + k)
  :=
2 begin
3   induction n with n' hn,
4     rw add_zero, -- Base Case: 0 + m + k = 0 + (m + k)
5     rw zero_add,
6     rw add_succ, -- Inductive Case: succ n' + m + k =
7                   succ n' + (m + k)
8     rw hn,       -- Use inductive hypothesis
9     rw add_succ, -- etc.
10  end
```



TIP: Press CMD+P to Verify.

?



Your Slide Title Here

- ❑ Your first point
- ❑ Your second point
- ❑ Your third point
- ❑ Your fourth point

☐ filename.ext ☐

Monaco

```
1  -- Replace this with your own code, pseudocode, or text.  
2  -- The box wraps long lines automatically.  
3  your_function(argument):  
4      return result
```



TIP: A short tip or callout goes here.

?



LEAN: Complete Proof Walkthrough

Lean Editor.pro

Monaco

```
1 theorem mul_comm (n m : N) : n * m = m * n :=
2 begin
3   induction n with n' hn,
4     rw zero_mul,      -- Base Case: 0 * m = m * 0
5     rw mul_zero,
6     rw succ_mul,      -- Inductive Case: succ n' * m = m * succ n'
7     rw hn,
8     rw mul_succ,
9 end
```



TIP: Press CMD+P to Verify.

?



LEAN: Course Roadmap

- Week 1: Propositional Logic
- Week 2: Natural Numbers & Induction
- Week 3: Lists & Recursion
- Week 4: Tactics Deep Dive
- Week 5: Building a Library
- Week 6: Final Project



TIP: No code today - just planning.

?



LEAN: Simplifying a Proof

Before

☐ proof_v1.lean ☒

Monaco

```
1 theorem add_zero'
2   (n : N) : n + 0 = n :=
3 begin
4   induction n with n' hn,
5     refl,
6     rw add_succ,
7     rw hn,
8 end
```

After

☐ proof_v2.lean ☒

Monaco

```
1 theorem add_zero'
2   (n : N) : n + 0 = n :=
3 by induction n;
4   simp [*, add_succ]
```



TIP: The simp tactic closes both cases at once.

?



A Note, Definition, or Quote

☐ Notes.txt ☐

Monaco

Replace this paragraph with your own text. The macbox environment gives you the same window chrome as macwindow, but typesets ordinary LaTeX - paragraphs, lists, tables, images - instead of code.

☐ Works with itemize, **bold**, and italic



TIP: Swap this box for a table, image, or blockquote.

?



LEAN: Simplifying a Proof

☐ proof_v1.lean ☐

Monaco

```
1 theorem add_zero'
2   (n : N) : n + 0 = n :=
3 begin
4   induction n with n' hn,
5     refl,
6     rw add_succ,
7     rw hn,
8 end
```



TIP: The simp tactic closes both cases at once.

?



By the Numbers

128

theorems proved

42

lemmas reused

6

weeks of work



TIP: Swap in your own three (or more) headline numbers.

?



References

Just for the citations [2][4][3][1]



References

- [1] Donald E. Knuth. “Literate Programming”. In: The Computer Journal 27.2 (1984), pp. 97–111.
- [2] Donald E. Knuth. The $\text{T}_{\text{E}}\text{X}$ Book. Addison-Wesley Professional, 1986.
- [3] Leslie Lamport. $\text{\LaTeX}$: a Document Preparation System. 2nd ed. Massachusetts: Addison Wesley, 1994.
- [4] Frank Mittelbach et al. The $\text{\LaTeX}$ Companion. 2nd ed. Addison-Wesley Professional, 2004.